# Calculus Skeleton Notes

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**Average Rates of Change**

**Goals:** Compute average rates of change, slopes of secant lines, and motivate the idea of limits
**Before Class Videos:** None today, starting next class

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**Example 1:** I left my house at time $t = 0$. 30 minutes later, I was at UVic which is 12 km away.

a) What was my average speed?

b) Sketch a possible graph

c) How could I approximate my speed at $t = 10$ minutes?

**Example 2:** What is average rate of change of $f(x) = x^2$ on $[2, 4]$?

**Example 3:** Find the equation of the line through $(x_1, y_1)$ and $(x_2, y_2)$.

**Example 4:** Consider the graph of a function $f(x)$. How do I write an equation for the secant line through $(x_1, f(x_1))$ and $(x_2, f(x_2))$?

**Big Idea:** As $x_2$ gets closer to $x_1$ the slope of the secant line approaches the slope of the tangent line at $x_1$. We need to investigate and define this limiting process more precisely. This DESMOS visualizes the idea: `https://www.desmos.com/calculator/oi7poyhodr`

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**Suggested after class problems** from APEX: None today, but review the Course Outline on Brightspace

## 1.1 - Limits

**Goals:** Describe limits graphically, numerically, and algebraically.
**Before Class Videos:** 1.1.1 and 1.1.2

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**Example 1:** Consider these three functions:

$$f(x) = x + 1, \qquad g(x) = \frac{x^2 - 1}{x - 1} = \frac{(x+1)(x-1)}{x-1}, \qquad h(x) = \begin{cases} x + 1, & x \neq 1, \\ 3, & x = 1. \end{cases}$$

How are these similar or different?

**Definition:**
$\lim_{x \to a} f(x) = L$ if $f(x)$ can be made arbitrarily close to $L$ for all values of $x$ sufficiently close to $a$.

**Example 2:** Compute the limit as $x \to 1$ of $f(x)$, $g(x)$, and $h(x)$.

**Example 3:** Use a table of values to estimate $\lim_{x \to 1}(x + 1)$.

**Example 4:** Use a table of values to estimate $\lim_{x \to 0} \frac{1}{x}$.

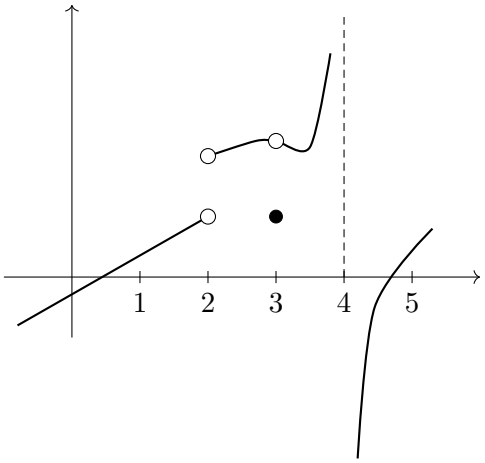
**Example 5:** Investigate

$$f(x) = \begin{cases} x, & x < 1, \\ x + 1, & x \geq 1. \end{cases}$$

**Definition:**

$\lim_{x \to a^+} f(x)$ means $f(x)$ gets arbitrarily close to $L$ for all $x$ sufficiently close to $a$ from the right, i.e. $x > a$.

**Example 6:** Use the graph to evaluate the indicated limits.




$\lim_{x\to 2^-} f(x) =$ _____, $\lim_{x\to 2^+} f(x) =$ _____, $\lim_{x\to 2} f(x) =$ _____,

$\lim_{x\to 3^-} f(x) =$ _____, $\lim_{x\to 3^+} f(x) =$ _____, $\lim_{x\to 3} f(x) =$ _____,

$\lim_{x\to 4^-} f(x) =$ _____, $\lim_{x\to 4^+} f(x) =$ _____, $\lim_{x\to 4} f(x) =$ _____.

**Example 7:** Investigate limits of

$$f(x) = \sin\left(\frac{1}{x}\right)$$

**Example 8:** Graphically investigate limits of $\frac{x^2-1}{x^2-x-2}$

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**Suggested after class problems** from APEX: Section 1.1 problems 9,11,15

## 1.3 - Finding Limits Analytically

**Goals:** Evaluate limits using limit laws, algebra, and the Squeeze Theorem.
**Before Class Videos:** 1.3.1–1.3.3

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**Definition:**

If $f(x)$ gets arbitrarily close to $L$ for all $x$ sufficiently close to $a$, then

$$\lim_{x \to a} f(x) = L.$$

For many "nice" functions, evaluating a limit just means plugging in the number. We'll investigate what functions are "nice" in section 1.5 on Continuity. For example:

**Example 1:**

- $\lim_{x \to 2} x^3 =$
- $\lim_{x \to \pi} \sin(x) =$
- $\lim_{x \to e} \ln(x) =$
- $\lim_{x \to 4} \sqrt{x} =$

**Theorem:**

If $L$, $M$, $c$, and $k$ are real numbers and

$$\lim_{x \to c} f(x) = L \qquad \text{and} \qquad \lim_{x \to c} g(x) = M,$$

then

| | |
|---|---|
| 1. Sum Rule: | $\lim_{x \to c}[f(x) + g(x)] = L + M$ |
| 2. Difference Rule: | $\lim_{x \to c}[f(x) - g(x)] = L - M$ |
| 3. Constant Multiple Rule: | $\lim_{x \to c}[kf(x)] = kL$ |
| 4. Product Rule: | $\lim_{x \to c}[f(x)g(x)] = LM$ |
| 5. Quotient Rule: | $\lim_{x \to c} \dfrac{f(x)}{g(x)} = \dfrac{L}{M}, \quad M \neq 0$ |
| 6. Power Rule: | $\lim_{x \to c}[f(x)]^n = L^n, \quad n$ a positive integer |
| 7. Root Rule: | $\lim_{x \to c} \sqrt[n]{f(x)} = \sqrt[n]{L} = L^{1/n}, \quad n$ a positive integer |

If $n$ is even in the Root Rule, assume that $f(x) \geq 0$ on an interval containing $c$.

**Example 2:** Evaluate

$$\lim_{x\to 2} \frac{x^2 - 2x}{x + 2}.$$

**Example 3:** Come up with functions $f$ and $g$ such that

$$\lim_{x\to 3} f(x)g(x) \neq \left[\lim_{x\to 3} f(x)\right] \left[\lim_{x\to 3} g(x)\right].$$

**Four Algebraic Tricks**

**Example 4:**

$$\lim_{x\to 2} \frac{x^2 - 2x}{x - 2}.$$

**Example 5:**

$$\lim_{x\to 0} \frac{(x-2)^2 - 4}{x}.$$

**Example 6:**

$$\lim_{x\to 9} \frac{\sqrt{x}-3}{x-9}.$$

**Example 7:**

$$\lim_{x\to 2} \frac{\frac{1}{2}-\frac{1}{x}}{x-2}.$$

**Example 8:** Evaluate

$$\lim_{x\to 4} \frac{4-x}{5-\sqrt{x^2+9}}.$$

**Theorem:**

**Squeeze Theorem.** If $f(x) \leq g(x) \leq h(x)$ for all $x$ in an interval containing $a$ (except possibly at $a$), and

$$\lim_{x \to a} f(x) = \lim_{x \to a} h(x) = L,$$

then

$$\lim_{x \to a} g(x) = L.$$

**Example 9:** Evaluate

$$\lim_{x \to 0} x^2 \sin(1/x^2).$$

**Example 10:** Suppose

$$-x^2 + 4x + 1 \leq g(x) \leq -2x + 10$$

for all $x$. For what value of $a$ can you determine $\lim_{x \to a} g(x)$? What is the limit?

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**Suggested after class problems** from APEX: Section 1.3 problems 8,11,29,36

## 1.5 - Continuity

**Goals:** Determine where a function is continuous, build continuous piecewise functions, and apply the Intermediate Value Theorem.
**Before Class Videos:** Section 1.5 videos 1 and 3

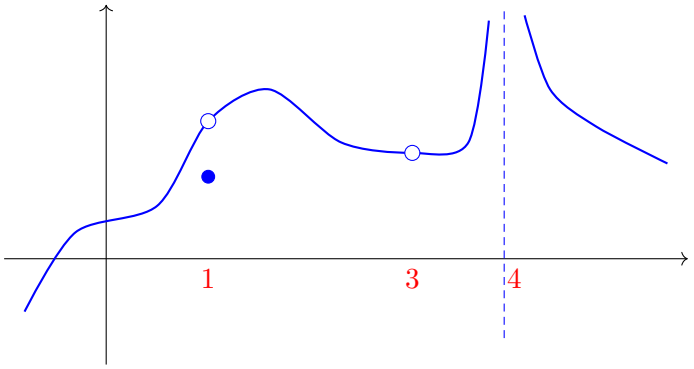
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**Definition:**

A function $f$ is **continuous at** $c$ if all three of the following hold:

1. $f(c)$ exists,
2. $\lim_{x \to c} f(x)$ exists, and
3. $\lim_{x \to c} f(x) = f(c)$.

**Example 1:** Identify the type of discontinuity at each marked $x$-value.




**Example 2:** For what values of $a$ is

$$f(x) = \begin{cases} x + 2, & x \leq 1, \\ ax^2, & x > 1 \end{cases}$$

continuous everywhere?

**Example 3:** For what values of $a$ and $b$ is

$$f(x) = \begin{cases} -2, & x \leq -1, \\ ax - b, & -1 < x < 1, \\ 3, & x \geq 1 \end{cases}$$

continuous everywhere?

### Combining Continuous Functions

**Theorem:**

If $f$ and $g$ are continuous at $c$, then $f+g$, $f-g$, constant multiples of $f$, and $fg$ are continuous at $c$. The quotient $f/g$ is also continuous at $c$ provided $g(c) \neq 0$.

Let's justify the sum rule from the limit laws:

### Composition of Functions

**Example 4:** Compute the domain and range of $\sin\sqrt{x}$

**Theorem:**

Suppose $f$ is continuous at $a$ and $g$ is continuous at $f(a)$. Then the composition $g \circ f$ is continuous at $a$, and

$$\lim_{x \to a} g(f(x)) = g(f(a)).$$

**Example 5:** On what intervals is $\sin\left(\frac{x-2}{x^2+5x+7}\right)$ continuous?

**Example 6:** Evaluate

$$\lim_{x \to 1} \frac{\sin(\pi x/6)e^{x^3} + x}{x^2 + 2}.$$

**Theorem:**

**Intermediate Value Theorem.** If $f$ is continuous on $[a, b]$ and $y_0$ is between $f(a)$ and $f(b)$, then there is some $c \in [a, b]$ such that

$$f(c) = y_0.$$

Sketch:

**Example 7:** Sketch examples where the assumptions of the Intermediate Value Theorem are not satisfied.

**Example 8:** Show that

$$x^3 - 3x - 1 = 0$$

has a solution in $[0, 2]$.

**Example 9:** Use the intermediate value theorem to show that at some time a baby's height equalled their child's height.

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**Suggested after class problems** from APEX: Section 1.5 problems 21,24,35

## 1.6 - Limits Involving Infinity

**Goals:** Investigate the special $\sin(x)/x$ limit, compute limits whose values are infinity, as well as limits at infinity

**Before Class Videos:** 1.6.1-1.6.3

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**Example 1:** Explore $\lim_{x \to 0} \dfrac{\sin x}{x}$.

**Example 2:** Compute $\lim_{x \to 0} \dfrac{\sin(5x)}{\sin(3x)}$.

**Example 3:** Explore

$$f(x) = \frac{x}{x-2}$$

and find all asymptotes.

**Definition:**

$\lim_{x \to a} f(x) = \infty$ if $f(x)$ can be made arbitrarily large for all values of $x$ sufficiently close to $a$.
*Note: A limit being infinite is a particular example of a limit not existing.*

**Definition:**

$\lim_{x \to \infty} f(x) = L$ if $f(x)$ can be made arbitrarily close to $L$ for all values of $x$ sufficiently large.

**Example 4:**

$$\lim_{x \to \infty} \sin(1/x)$$

**Example 5:**

$$\lim_{x \to \infty} x \sin(1/x)$$

**Example 6:** Find all Horizontal Asymptotes for

$$\frac{2x^3 + 3x}{x^4 - x^2}.$$

**Example 7:** Find all Horizontal Asymptotes for

$$\frac{\sqrt{x^6+2x}}{x^3+x^{1/3}}.$$

**Example 8:** Find VA's; HA for $e^{1/x}$; graph it!

**Example 9:** Give example of a function with two HAs, $y=0$ and $y=2$, continuous everywhere. Hint: $\arctan(x)$.

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**Suggested after class problems** from APEX: Section 1.4 #40 and Section 1.6 #10, 19,27

## 2.1 - Derivatives

**Goals:** Define the Derivative and apply it to initial examples
**Before Class Videos:** 2.1.1, 2.1.2

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**Example 1:** Recall the velocity problem from the first day of class. If Trefor has some distance function $f(t)$, how fast is he going at 10 minutes? Let's explore graphically and numerically:

**Definition:**
The derivative of $f(x)$ at $c$ is given by

$$f'(c) = \lim_{h \to 0} \frac{f(c+h) - f(c)}{h}$$

provided this limit exists. If it does, we say $f$ is differentiable at $c$.

**Example 2:** Compute the derivative of $f(x) = x^2$ at $x = 2$.

**Definition:**
The derivative of $f(x)$ on an open interval $I$ is the function $f'(x)$ given by

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

provided the limit exists for all $x$ in $I$. If it does we say $f(x)$ is differentiable on $I$.

**Example 3:** $f(x) = 1/x^2$.

**Example 4:** $f(x) = \sqrt{x}$.

**Example 5:** Let's explore if the following functions are differentiable at $x = 0$:

a)

$$f(x) = \begin{cases} 2, & x \leq 0, \\ 1, & x > 0. \end{cases}$$

b) $f(x) = |x|$

c) $f(x) = x^{2/3}$

d) $f(x) = x^{1/3}$

---

**Suggested after class problems** from APEX: Section 2.1: 9, 17, 24

## 2.2 - Derivatives

**Goals:** A shorter than normal worksheet with a few extras on derivatives
**Before Class Videos:** Optional tricky example from last class's content: 2.1.4

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**Definition:**
The derivative of $f(x)$ at $c$ is given by

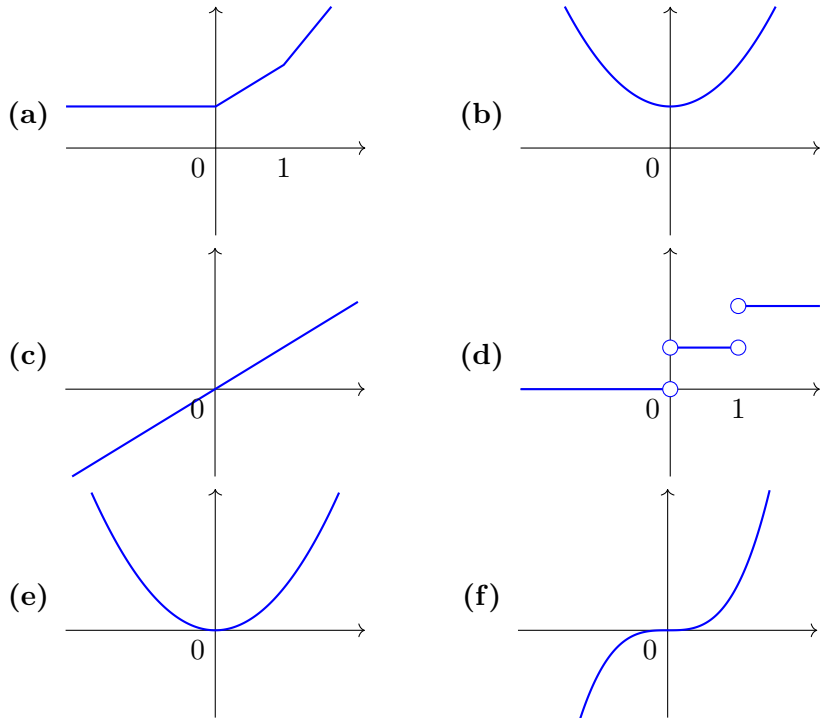
$$f'(c) = \lim_{h \to 0} \frac{f(c+h) - f(c)}{h}$$

provided this limit exists. If it does, we say $f$ is differentiable at $c$.

**Definition:**
An alternate formula for the derivative of $f(x)$ at $c$ is

$$f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c}.$$

**Example 1:** The following six graphs consists of three pairs of functions and their derivative. Match them up.




**Theorem:**
Differentiable $\implies$ Continuous.

Note: This worksheet is deliberately short. I anticipate this class being a bit behind and finishing up prior material. If I am wrong I will edit it and add to it. The next worksheet is a more normal length.

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**Suggested after class problems** from APEX: Section 2.2 15,17

## 2.3 - Differentiation Rules

**Goals:** State and apply Derivative Rules
**Before Class Videos:** 2.3.1

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**Theorem (Power Rule):**

- $\frac{d}{dx}(c) = 0$ where $c$ is a constant.
- $\frac{d}{dx}(x^n) = nx^{n-1}$ where $n \neq 0$

**Theorem:**

If $f(x)$ and $g(x)$ are differentiable at $x = a$, then

1. **Sum/Difference Rule**:

$$\frac{d}{dx}[f(x) \pm g(x)] = \frac{d}{dx}f(x) \pm \frac{d}{dx}g(x)$$

2. **Constant Multiple Rule**:

$$\frac{d}{dx}[cf(x)] = c\frac{d}{dx}f(x)$$

**Example 1:** Compute the derivative of $f(x) = 3x^4 - 5\sqrt{x} + 6x^{3/2} - \frac{7}{x^2} + 1$.

**Proof of Theorem:**

**Definition:**
The Point-Slope formula for the equation of the straight line through $(x_0, y_0)$ with slope $m$ is

$$y = y_0 + m(x - x_0)$$

**Example 2:** Find the equation of tangent line to a differentiable function $f(x)$ at the point $x_0$.

**Example 3:** Find the equation of the tangent line to the function $f(x) = 2x^2 + 1$ at the point $(1, 3)$.

**Common Derivatives:**

**Example 4:** Explore the graphs of the following functions to discover their derivatives.

- $\frac{d}{dx}(e^x) =$

- $\frac{d}{dx}\ln(x) =$ for $x > 0$

- $\frac{d}{dx}\sin(x) =$

- $\frac{d}{dx}\cos(x) =$

**Higher Derivatives:** The derivative $f'(x)$ is itself a function we can take the derivative of. We call this the second derivative and might denote it by

$$f''(x) \qquad \text{or} \qquad \frac{d^2}{dx^2}f(x).$$

**Example 5:** Compute $f''(x)$ when $f(x) = e^x + \ln(x)$.

**Example 6:** If $f(x) = \sin(x)$ compute

- $f'(x) =$
- $f''(x) =$
- $f'''(x) =$
- $f^{(4)}(x) =$
- $f^{(2026)}(x) =$

**Example 7:** If $d(t)$ represents the position of a particle as a function of time, interpret $d'(t)$ and $d''(t)$.

**Example 8:** If $v(t)$ represents Trefor's total YouTube views after $t$ days, interpret $v'(t)$ and $v''(t)$.

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**Suggested after class problems** from APEX: Section 2.3 16, 21, 34 (tangent only)

## 2.4 - Product and Quotient Rule

**Goals:** Compute derivatives using the product and quotient Rules, including of trigonometric functions

**Before Class Videos:** 2.4.1 and 2.4.2

---

**Theorem (Derivative Rules):**

- $\frac{d}{dx}(c) = 0$ where $c$ is a constant.
- $\frac{d}{dx}(x^n) = nx^{n-1}$ where $n > 0$ is an integer.

If $f(x)$ and $g(x)$ are differentiable at $x = a$, then

- $\frac{d}{dx}[f(x) \pm g(x)] = \frac{d}{dx}f(x) \pm \frac{d}{dx}g(x)$
- $\frac{d}{dx}[cf(x)] = c\frac{d}{dx}f(x)$

Let's add two new rules to the list

**Theorem (Product and Quotient Rules):**

$$\frac{d}{dx}[fg] = f'g + fg'$$

$$\frac{d}{dx}\left[\frac{f}{g}\right] = \frac{f'g - fg'}{g^2}$$

**Example 1:** Compute the following derivatives

a)

$$\sqrt{x}e^x$$

b)

$$\frac{\sin(x)}{x^3}$$

c)

$$\frac{xe^x}{x^2+1}$$

**Example 2:** Let's explore the intution behind why the product rule is NOT just the product of the derivatives.

**Example 3:** Let's finally figure out what exactly $e^x$ is all about. We'll use the definition of the derivative to compute, for $a > 0$,

$$\frac{d}{dx}a^x$$

Let's explore

$$\lim_{h\to 0} \frac{a^h - 1}{h}$$

**Definition:**

The number $e$ is defined to be the value of $a$ such that $\lim_{h\to 0} \frac{a^h - 1}{h}$

**Right-Triangle Definitions**

$$\sin x = \frac{\text{opposite}}{\text{hypotenuse}} \quad \tan x = \frac{\text{opposite}}{\text{adjacent}} \quad \sec x = \frac{\text{hypotenuse}}{\text{adjacent}}$$

$$\cos x = \frac{\text{adjacent}}{\text{hypotenuse}} \quad \cot x = \frac{\text{adjacent}}{\text{opposite}} \quad \csc x = \frac{\text{hypotenuse}}{\text{opposite}}$$

**Rewriting with Sine and Cosine**

$$\sin x = \sin x \quad \tan x = \frac{\sin x}{\cos x} \quad \sec x = \frac{1}{\cos x}$$

$$\cos x = \cos x \quad \cot x = \frac{\cos x}{\sin x} \quad \csc x = \frac{1}{\sin x}$$

**Example 4:** Let's explore the graphs of these functions.

**Example 5:** Compute $\frac{d}{dx}\tan(x)$

**Example 6:** Compute $\frac{d}{dx}\sec(x)$

**Trigonometric Derivatives**

$$\frac{d}{dx}\sin x = \cos x \qquad \frac{d}{dx}\tan x = \sec^2 x \qquad \frac{d}{dx}\sec x = \sec x \tan x$$

$$\frac{d}{dx}\cos x = -\sin x \qquad \frac{d}{dx}\cot x = -\csc^2 x \qquad \frac{d}{dx}\csc x = -\csc x \cot x$$

---

**Suggested after class problems** from APEX: Section 2.4 7, 18, 26

## 2.5 - Chain Rule

**Goals:** Compute the derivative of compositions of functions
**Before Class Videos:** 2.5.1, 2.5.2

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**Example 1:** Consider $\sin(x^2)$. This is an example of a composition $f(g(x))$ or alternatively $f \circ g$.

**Example 2:** $f(x) = x \sin x$, $g(x) = \cos x + 1$. Write out:

- $f(g(x)) =$
- $g(f(x)) =$
- $f(f(x)) =$
- $g(g(x)) =$

**Theorem (Chain Rule):**
If $g$ differentiable at $x$ and $f$ differentiable at $g(x)$ then

$$\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x).$$

**Example 3:** $\dfrac{d}{dx} \sin(x^2 + 1)$

**Example 4:** $\dfrac{d}{dx} \sin^{30}(x)$

**Example 5:** $\dfrac{d}{dx} \cos(3e^x)$

**Example 6:** $\dfrac{d}{dx}\dfrac{1}{2x+1}$

**Example 7:** $\dfrac{d}{dx}\cos^2(x^3)$

**Example 8:** $\dfrac{d}{dx}\tan(x^2e^x)$.

**Leibniz Notation**: let $u = g(x)$,

$$\frac{d}{dx}[f(g(x))] = \frac{df}{du}\frac{du}{dx}.$$

**Example 9:** Car uses $6L/100km$. Cost is $1.5\$/L$. What is the rate of change in cost per $100km$?

**Example 10:** Gravel is dumped from a conveyor to form pile with

$$V(h) = \frac{\pi}{12}h^3.$$

At $5m$ tall, the pile is growing by 0.1 m/s. How fast is volume growing?

**Example 11:** Contrast graphically the slope of $f(x)$ and $f(2x)$ for an arbitrary function $f(x)$.

**Example 12:** Derive the Quotient Rule.

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**Suggested after class problems** from APEX: Section 2.5: 7,17,24

## 2.6 - Implicit Differentiation

**Goals:** Compute derivatives of functions define implicitly
**Before Class Videos:** 2.6.1

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**Example 1:** Explore $x^2 + y^2 = 1$. Where does it have vertical and horizontal tangents?

**Big Idea:** Treat $y = y(x)$ when $x$ is independent and $y$ is dependent.

**Example 2:** Find $\dfrac{dy}{dx}$ implicitly for $\sin(xy) + x = y^2 \cos x$.

**Example 3:** Compute $\dfrac{d^2y}{dx^2}$ for $x^2 + y^2 = 4$.

**Example 4:** Now explore $x^3 + y^3 - 9xy = 0$. Find all vertical and horizontal tangents.

**Logarithmic Differentiation:** Recall we previously learned that $\frac{d}{dx}ln(x) = \frac{1}{x}$.

**Example 5:** $y = x^x$.

**Example 6:**

$$y = \frac{(x+2)^3(x-1)^{1/3}}{(x+7)^2}.$$

**Example 7:**

$$y = x^{\cos x} \cdot (\cos x)^x.$$

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**Suggested after class problems** from APEX: Section 2.6: 15,17,40

## 2.7 - Inverses and Inverse Trig

**Goals:** Compute the Derivatives of Inverse functions and Inverse Trig functions.
**Before Class Videos:** 2.7.1

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*Note: This worksheet combined some content from inverses in general and trignometric inverses. The former was covered on Oct 7th by Josh Manzer using his own worksheet and will thus not be covered extensively by Trefor, who will focus on the inverse trig for Friday's Class.*

**Definition:**
For a function $f(x)$, let's define $f^{-1}(x)$ so

$$f(f^{-1}(x)) = f^{-1}(f(x)) = x.$$

**Example 1:** $y = 2x + 1$

**Example 2:** $y = x^2$

**Example 3:** $\dfrac{d}{dx} \ln x$

**Example 4:** $\dfrac{d}{dx}a^x$

**Example 5:** Let's derive a general formula for the inverse of a function.

**Example 6:** $\tan^{-1}(\tan x) = x = \tan(\tan^{-1} x)$.Try evaluating these at $x = \pi$.

**Example 7:** $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$

**Example 8:** $\sin^{-1}\left(-\frac{1}{\sqrt{3}}\right)$

**Example 9:** Plot the graph of $\arcsin(x)$ by comparing to the graph of $\sin(x)$.

**Example 10:** Compute $\dfrac{d}{dx}\arcsin(x)$.

**Inverse Trigonometric Functions**

| Function | Derivative | Domain | Range |
|---|---|---|---|
| $\arcsin x$ | $\dfrac{1}{\sqrt{1-x^2}}$ | $[-1,1]$ | $[-\pi/2,\pi/2]$ |
| $\arccos x$ | $-\dfrac{1}{\sqrt{1-x^2}}$ | $[-1,1]$ | $[0,\pi]$ |
| $\arctan x$ | $\dfrac{1}{1+x^2}$ | $\mathbb{R}$ | $(-\pi/2,\pi/2)$ |
| $\operatorname{arccot} x$ | $-\dfrac{1}{1+x^2}$ | $\mathbb{R}$ | $(0,\pi)$ |
| $\operatorname{arcsec} x$ | $\dfrac{1}{\lvert x\rvert\sqrt{x^2-1}}$ | $(-\infty,-1]\cup[1,\infty)$ | $[0,\pi/2)\cup(\pi/2,\pi]$ |
| $\operatorname{arccsc} x$ | $-\dfrac{1}{\lvert x\rvert\sqrt{x^2-1}}$ | $(-\infty,-1]\cup[1,\infty)$ | $[-\pi/2,0)\cup(0,\pi/2]$ |

The derivatives of $\arcsin x$ and $\arccos x$ hold for $|x|<1$; those of $\operatorname{arcsec} x$ and $\operatorname{arccsc} x$ hold for $|x|>1$.

**Example 11:** $\dfrac{d}{dx}\sin^{-1}(3x^2+1)$

**Example 12:** Let's do one more derivativion, this time of $y = \sec^{-1}(x)$.

**Example 13:** $\dfrac{d}{dx}[\ln(x^2+1) + x\tan^{-1}(2x)]$

**Example 14:** For $\tan^{-1} x + \sin^{-1} y = \frac{\pi}{4}$ find $\left.\dfrac{dy}{dx}\right|_{(0,\frac{1}{\sqrt{2}})}$

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**Suggested after class problems** from APEX: Section 2.7: 9,13,16,23,29

## 4.2 - Related Rates

**Goals:** Relate changing quantities by differentiating equations with respect to time.
**Before Class Videos:** 4.2.1

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**Example 1:** An inverted conical tank has radius 6 ft and height 8 ft. Water is pumped in at 4 $ft^3/min$. How fast is the water level rising when the water is 4 ft deep?

**Example 2:** A rocket rises vertically at 600 m/s. An observer stands 4000 m from the launch point. How fast is the angle of elevation changing when the rocket is 3000 m high?

**Method:**

**Example 3:** Two boats are 400 km apart horizontally. Boat A goes north at 30km/h while boat B goes south at 45 km/h. How fast is the distance between them changing after 2 hours?

**Example 4:** A point moves along the curve $y = x^2$, with $x > 0$. Find the rate of change of its distance from $(3, 0)$ in terms of $x$ and $dx/dt$.

**Example 5:** Suppose a bar of length $d$ leans against a wall and slides along the ground. If $x'(t)$, the rate of change of its horizontal position is constant, is $y'(t)$ constant?

**Example 6:** Suppose a $2m$ tall skateboarder is a distance $x$ from a $5m$ tall wall which has a light at the top. If $x'(t) = 2m/s$, how fast is the length of the shadow cast by the light increasing?

**Example 7:** A trough is 20 m long, with right triangular ends 5 m high and 5 m wide. Water flows in at 100 $m^3$/h. How fast is the water level rising when the water is 2 m deep?

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**Suggested after class problems** from APEX: Section 4.2 problems 3,5,7

## 3.1 - Extreme Values

**Goals:** Identify and find absolute and relative extrema, particularly on closed intervals.
**Before Class Videos:** 3.1.1, 3.1.2

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**Definition:**

Let $f$ be defined on an interval $I$ containing $c$.

1. $f(c)$ is the **absolute** (or global) maximum if $f(c) \geq f(x)$ for all $x$ in $I$.
2. $f(c)$ is the **absolute** (or global) minimum if $f(c) \leq f(x)$ for all $x$ in $I$.
3. $f(c)$ is a **relative** (or local) maximum if $f(c) \geq f(x)$ for all $x$ in some interval about $c$ within $I$.
4. $f(c)$ is a **relative** (or local) minimum if $f(c) \leq f(x)$ for all $x$ in some interval about $c$ within $I$.

**Theorem (Extreme Value Theorem):**

On a closed interval $[a, b]$, a continuous function must attain an absolute maximum and absolute minimum in the interval.

**Example 1:** Sketch examples where the assumption is not satisfied and the conclusion of the Extreme Value Theorem is false.

**Example 2:** What are the maximum and minimum values of $f(x) = x^2 - 6x + 7$ on $[2, 5]$.

**Definition:**
A **critical number** of $f$ is a number $c$ in the domain of $f$ such that either $f'(c) = 0$ or $f'(c)$ does not exist. A **critical point** is a pair $(c, f(c))$ when $c$ is a critical number.

**Example 3:** Find the absolute maximum and minimum of $f(x) = x^{2/3}$ on $[-1, 1]$.

**Extreme Value Method for Closed Intervals:**

1. Find all critical numbers: $f'(x) = 0$ or DNE.
2. Evaluate $f(x)$ at each critical number (in the interval) and endpoints of the interval
3. Pick biggest/smallest.

**Example 4:** Find the absolute max and min of $f(x) = x^4 - 8x^2$ on $[-1, 3]$.

**Example 5:** $f(x) = x^{1/3}$ on $[-1, 1]$.

**Theorem:**
If $f$ has a max at $c$ and $f'(c)$ exists, then $f'(c) = 0$.

**Example 6:** $f(x) = \ln(x+1) + \tan x$ on $(-1, 1]$.

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**Suggested after class problems** from APEX: Section 3.1: 7,10,17,19

## 3.2 - Mean Value Theorem

**Goals:** State and Apply Rolle's Theorem and the Mean Value Theorem
**Before Class Videos:** 3.2.1

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*Note: This was originally posted for last friday, but we never began it so it is relabelled as Oct 20th notes*

**Theorem (Rolle's Theorem):**
If $f$ is continuous on $[a, b]$, differentiable on $(a, b)$, and $f(a) = f(b)$, then there exists $c \in (a, b)$ so $f'(c) = 0$.

Proof:

**Theorem (Mean Value Theorem):**
Suppose $f(x)$ is continuous on $[a, b]$ anddifferentiable on $(a, b)$.
Then there exists $c$ in $(a, b)$ so

$$\frac{f(b) - f(a)}{b - a} = f'(c).$$

**Example 1:** $y = \sqrt{x-2}$ on $[2, 6]$.

**Example 2:** Suppose $f(1) = 10$ and $f'(x) \geq 2$ on $1 \leq x \leq 4$. What is a minimum possible value of $f(4)$?

**Example 3:** $f(x) = x^3 + 3x + 1$. Show exactly one root.

**Theorem:**
Suppose $f'(x) = 0$ on $[a, b]$. Then $f(x)$ is constant.

**Corollary:** Suppose $f'(x) = g'(x)$ on $(a, b)$. Then $f(x) = g(x) + C$ for some constant.

**Example 4:** If $f'(x) = x^2$, what is $f(x)$?

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**Suggested after class problems** from APEX: Section 3.2: 1,5,11

## 3.3 - Increasing Functions

**Goals:** Connect properties of the derivative to the idea of increasing functions and extrema
**Before Class Videos:** No video for this section; consult Section 3.3 of APEX Calculus to review.

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**Definition:**
$f(x)$ increasing on interval $I$ if for all $a < b$ in $I$ $f(a) < f(b)$ in $I$.

**Definition:**
$f(x)$ decreasing on interval $I$ if for all $a < b$ in $I$ $f(a) > f(b)$ in $I$.

**Theorem:**
If $f'(x) > 0$ on $[a, b]$, then $f(x)$ increasing on $[a, b]$.

**Example 1:** Decide the intervals on which $f(x) = 2x^3 - 3x^2 - 12x$ is increasing and decreasing. Sketch a very loose graph of the function.

**Theorem (1st Derivative Test):**

Let $c$ be a critical number of $f$, and $f$ both continuous at $c$ and differentiable near $c$.

1. $f'(x)$ changes sign from $+$ to $-$ then $f(c)$ is a relative max.
2. $f'(x)$ changes sign from $-$ to $+$ then $f(c)$ is a relative min.
3. If there is no change in sign, then $f(c)$ is not a relative extremum

**Example 2:** Classify all critical numbers of

$$f'(x) = \frac{e^{-2x}}{20}(x-1)(x+1)^2.$$

**Example 3:** Classify all critical numbers of

$$g'(x) = \frac{(x+4)(x-1)^2}{x-2},$$

**Example 4:** Sketch graph of function such that:

- V.A. at $x = 1$;
- $f'(x) < 0$ on $(1, \infty)$;
- $f'(x) > 0$ on $(-1, 1)$;
- minimum at $x = -1$;
- $\lim_{x \to -\infty} f(x) = 0$;
- continuous elsewhere.

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**Suggested after class problems** from APEX: Section 3.3: 15,21